

Design Approach to Mach number 0.5 Low Speed Subsonic Wind Tunnel

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ABSTRACT

Wind tunnel is a device used to test aero models, automobiles and building structures. It is used to give appropriate shape to the structure. We calculated aerodynamic characteristics analytically for different models whereas wind tunnel can give practical values and we wanted to know how this can be designed. We decided to design an open circuit low speed subsonic suction type wind tunnel. Then, for a particular wind tunnel design the test section dimensions were defined according to the model size and air speed requirement. Parameters and dimensions at different sections were found by standardized equations like continuity and isentropic flow equations. We calculated boundary layer thickness of on the test section wall for designed velocity to check the impact of it on the flow in the test section of our designed wind tunnel. . Also, we calculate the boundary layer thickness in different velocities to ensure that the flow over the testing object in test section is stable for all possible velocities.

KEYWORDS: *Contraction Cone, Test Section, Diffuser, Honeycomb Structure, Fan*

NOMENCLATURE

A_R	Diffuser to test section area ratio	l_{11}	Length of honeycomb
A_1	Area of cross section of nozzle inlet	l_{12}	Length of contraction(nozzle)
A_2	Test section cross section area	l_{13}	Length of settling chamber
A_3	Cross section area of diffuser exit	k_d	Diffuser loss co-efficient
b_h	Breadth of honeycomb cell	k_{ex}	Expansion loss co-efficient
b_1	Breadth of nozzle cross section	k_f	Friction loss co-efficient
b_2	Breadth of test section cross section	$\dot{m}_1$	Mass flow rate in nozzle
d_3	Diffuser End cross section diameter	$\dot{m}_2$	Test section mass flow rate
h_h	Height of honeycomb cell	$\dot{m}_3$	Mass flow rate in diffuser section
h_1	Height of nozzle cross section	p_{atm}	Atmospheric air pressure
h_2	Height of test section cross section	P_m	Motor power
l_d	Diffuser section length	p_1	Pressure at inlet of nozzle
l_h	Honeycomb structure length	p_2	Test section inlet pressure
l_1	Total length of nozzle section	p_3	Pressure at diffuser exit
l_2	Test section total length	r_3	Radius of diffuser exit cross section
l_3	Diffuser length		

v_1	Velocity at nozzle Inlet
v_2	Velocity of flow in test section
v_3	Flow velocity of diffuser exit
θ	Diffuser angle
ρ_{atm}	Atmospheric air density

ρ_1	Density at nozzle inlet
ρ_2	Test section inlet density
ρ_3	Flow density at diffuser exit

INTRODUCTION

Wind tunnels were first proposed as a means of studying vehicles (primarily airplanes) in free flight. The wind tunnel was envisioned as a means of reversing the usual paradigm: instead of the air's standing still and the aircraft moving at speed through it, the same effect would be obtained if the aircraft stood still and the air moved at speed past it. In that way a stationary observer could study the aircraft in action, and could measure the aerodynamic forces being imposed on the aircraft.

Later, wind tunnel study came into its own: the effects of wind on manmade structures or objects needed to be studied, when buildings became tall enough to present large surfaces to the wind, and the resulting forces had to be resisted by the building's internal structure. Determining such forces was required before building codes could specify the required strength of such buildings. Wind-tunnel testing was applied to automobiles, not so much to determine aerodynamic forces but more to determine ways to reduce the power required to move the vehicle on roadways at a given speed.

The airflow created by the fans that is entering the tunnel is itself highly turbulent due to the fan blade motion (when the fan is blowing air into the test section - when it is sucking air out of the test section downstream, the fan-blade turbulence is not a factor), and so is not directly useful for accurate measurements. The air moving through the tunnel needs to be relatively turbulence-free and laminar. To correct this problem, closely-spaced vertical and horizontal air vanes are used to smooth out the turbulent airflow before reaching the subject of the testing.

Due to the effects of viscosity, the cross-section of a wind tunnel is typically circular rather than square, because there will be greater flow constriction in the corners of a square tunnel that can make the flow turbulent. A circular tunnel provides a smoother flow. The inside facing of the tunnel is typically as smooth as possible, to reduce surface drag and turbulence that could impact the accuracy of the testing. Even smooth walls induce some drag into the airflow, and so the object being tested is usually kept near the center of the tunnel, with an empty buffer zone between the object and the tunnel walls. There are correction factors to relate wind tunnel test results to open-air results. One can look for following results from the wing tunnel experiment:

Pressure measurements

Force and moment measurements

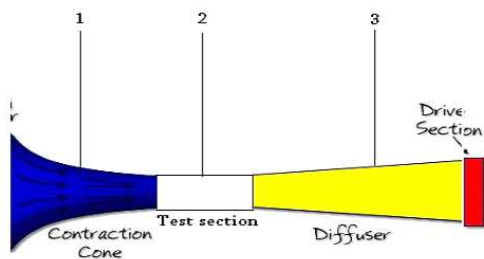
Flow visualization

We used to make air-craft models and gliders in our Lab in Undergraduate Study. We also mounted engine on couple of them and flew the models with R/C remote. We also did wind tunnel practical as our lab work. We were excited and curious about designing a wind tunnel that can test the models made by us. The test section, speed requirement and some other basic specification were taken considering this aim.

PARTS OF WIND TUNNEL

Most designs feature each of the five components described below. The overall design creates high-speed, low-turbulence airflow through the test section and allows researchers to measure the resulting forces on the model being tested. Importance of different parts is as below:

Settling Chamber - The purpose of the settling chamber is to straighten the airflow.



Contraction Cone - The contraction cone takes a large volume of low-velocity air and reduces it to a small volume of high velocity air without creating turbulence.

Test Section - The test section is where the test article and sensors are placed.

Diffuser - The diffuser slows the speed of airflow in the wind tunnel.

Drive Section - The drive section provides the force that causes the air to move through the wind tunnel.

Wind tunnels can be open or closed-loop. The design influences whether or not the air is recirculated in the tunnel.

WIND TUNNEL DESIGN

We set the requirements that will allow us to design a wind tunnel that can be used to test models at low subsonic speeds. The test section was kept to be $0.75\text{m} \times 0.75\text{m} \times 1\text{m}$. Maximum airspeed in the test section is 0.5M . Now we know the speed requirement in the test section. We also know the atmospheric pressure, density and temperature outside the wind tunnel. Notation for different parts of wind tunnel is taken as Contraction cone -1, Test Section-2 and Diffuser-3. Using continuity and isentropic equations shown below

allows finding atmospheric parameters at different locations in the wind tunnel.

Continuity Equation:

$$\dot{m} = \rho AV = \text{constant} \quad \text{-----}(1)$$

Isentropic Flow Equation:

Density,

$$\frac{\rho_0}{\rho} = \left[1 + \frac{\gamma-1}{2} M^2 \right]^{\frac{1}{\gamma-1}} \quad \text{-----}(2)$$

Pressure,

$$\frac{p_0}{p} = \left[1 + \frac{\gamma-1}{2} M^2 \right]^{\frac{\gamma}{\gamma-1}} \quad \text{-----}(3)$$

Mach No,

$$M = \frac{v}{a} \quad \text{-----}(4)$$

Assumptions

Inlet velocity from the contraction cone is assumed to be 0.02M which is almost zero.

Atmospheric air is considered as a perfect gas.

The air flow is isentropic.

Calculation

Test section area $A_2 = 0.75\text{m} \times 0.75\text{m}$

$$A_2 = 0.5625 \text{ m}^2$$

$$r_2 = 0.42\text{m}$$

Speed in the test section is 0.5M

$$M = \frac{v}{a} \quad ; \quad v = 170 \text{ m/s}$$

Contraction cone, Atmospheric flow parameters at contraction cone (section 1)

$$k_{ex(circle)} = \begin{cases} 0.1033 - 0.02389\theta & \text{for } 0^\circ < \theta < 1.5^\circ \\ 0.1709 - 0.1170\theta + 0.03260\theta^2 + 0.001078\theta^3 \\ -0.0009076\theta^4 - 0.00001331\theta^5 + 0.00001345\theta^6 & \text{for } 1.5^\circ \leq \theta \leq 5^\circ \\ -0.09661 + 0.04672\theta & \text{for } 5^\circ < \theta \end{cases}$$

After using different values of θ , the best value found

$$k_f = \left(1 - \frac{1}{A_R^2}\right) \frac{f}{8 \sin \theta}$$

$$k_f = 0.25$$

$$k_d = 0.3118$$

From isentropic flow table for 0.5M (Yahya, H.),

$$\frac{A_t}{A_*} = 1.34$$

$$\frac{A_t}{A_*} = \frac{A_t}{A_e} \times \frac{A_e}{A_*}$$

$$\frac{A_t}{A_e} = \frac{\text{area of test section}}{\text{area of diffuser}} = \frac{0.5625}{1.71}$$

$$\frac{A_e}{A_*} = \frac{A_t}{A_*} \times \frac{A_e}{A_t}$$

$$= 1.34 \times 3$$

$$\frac{A_e}{A_*} = 4.02$$

Isentropic flow properties table suggest that mach no is the diffuser section lies between 0.14M and 0.16M. For 0.147M interpolation method gives velocity of 50m/s. 50m/s needed to be produced at the diffuser section with the suction fan. Force and Power required for that are:

$$\Delta p = p_{atm} - p_3$$

$$\Delta p = 0.015 \times 10^5 \text{ N/m}^2 \quad ; \quad F = \Delta p \cdot A_3 = 12.9$$

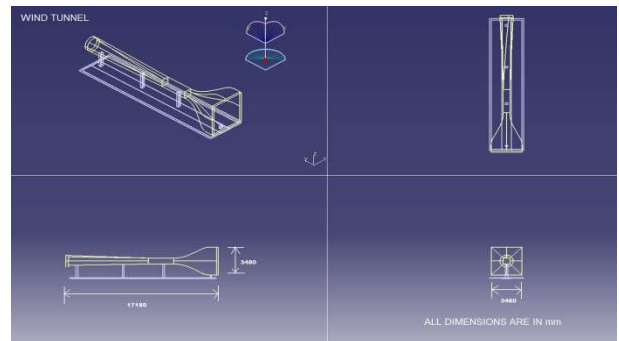
$$\text{KN}; \quad P = F \cdot V_3 = 171.12 \text{ hp}$$

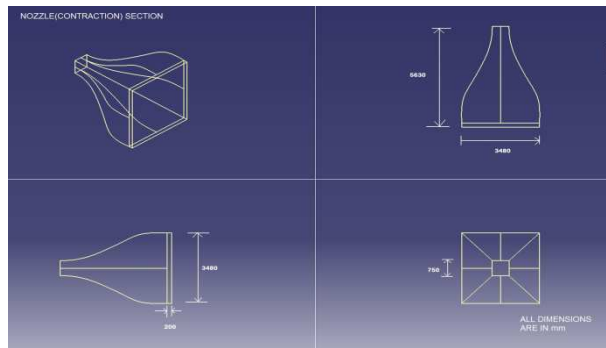
Result Table

was 2° which give minimum expansion loss co-efficient. With diffuser angle of 2° , diffuser length is 9m. Trigonometric calculation gives 0.74m radius for diffuser section. Hence, diffuser section area is 1.72m^2 . Diffuser to test section area ratio is 3.05. From historical data it is stated that the diffuser to test section area ratio is typically 2 to 3.

RESULTS

	Symbol	Magnitude	Unit
Test section	ρ_2	1.088	kg/m^3
	p_2	0.8541×10^5	N/m^2
Contraction cone	v_1	6.8	m/s
	A_1	12.15	m^2
	$h_1 = b_1$	3.48	m
	l_1	4.8	m
Honeycomb Structure	A_h	4.86×10^{-4}	m^2
	$h_h = b_h$	2.20×10^{-2}	m
	l_h	0.2	m
Diffuser	l_3	9	M
	θ	2°	Deg
	A_3	1.72	m^2
	r_3	0.74	M
	k_{ex}	0.061	-
	k_f	0.25	-
	k_d	0.311	-
	v_3	49.48	m/s
	p_3	0.99×10^5	N/m^2
Fan	P	171.12	hp



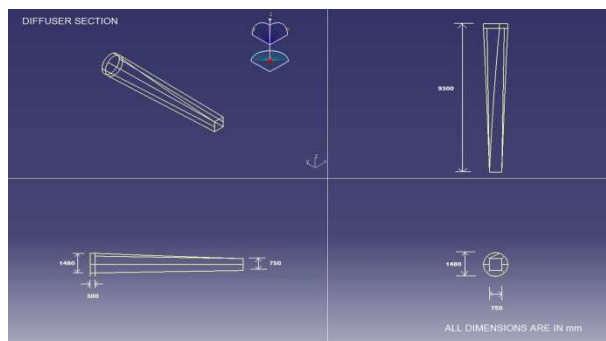
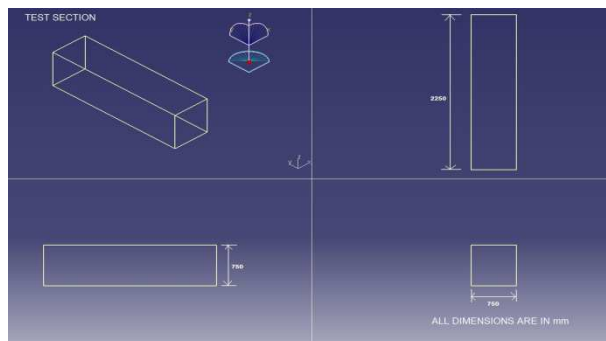


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CONCLUSION

Calculations at different stages and historical data confirmed that the wind tunnel design is correct. At the designed Mach number 0.5, corresponding to velocity of 170.7m/s, the boundary layer thickness ranges from 0.000048m to 0.0142m which is acceptable. So, there is no need to diverge the test section.

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